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ZF - Division A

Slip Control Software Design

Document ID	:	ARM-DDD-Arm_Roll_Over_Index
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REFERENCES

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1 ABSTRACT

This module calculates the Roll Over Index ([ROI](#)) and its activation threshold.

2 ACRONYMS

ARM Active Roll Mitigation

AyMod 'Modified' Lateral Acceleration, that tries to predict the Lateral Acceleration

ROI Roll Over Index

COG Center of Gravity

m Vehicle Mass

WSR Wheel Slip Regulator

3 OVERVIEW

This document describes the purpose and behavior of Arm_Roll_Over_Index.

3.1 DESIGN GOAL

The predicted lateral acceleration [AyMod](#) indicates the roll over propensity and should be considered as activation condition for [ARM](#). Furthermore the roll over propensity indicates the amount of intervention, that is necessary to avoid an impending vehicle roll over.

3.2 GENERAL REQUIREMENTS

1. [ARM](#) should prevent dynamic rollover.
2. [ARM](#) should prevent static rollover.

3.3 DETAILED REQUIREMENTS

Requirement ID	
Location	

3.4 SYSTEM MODE REQUIREMENTS

This module needs only to be processed, when **ARM** is allowed.

3.5 FLOW-CHARTS AND DIAGRAMS

The calculation of **ROI** is presented in Fig. 3.1 below.

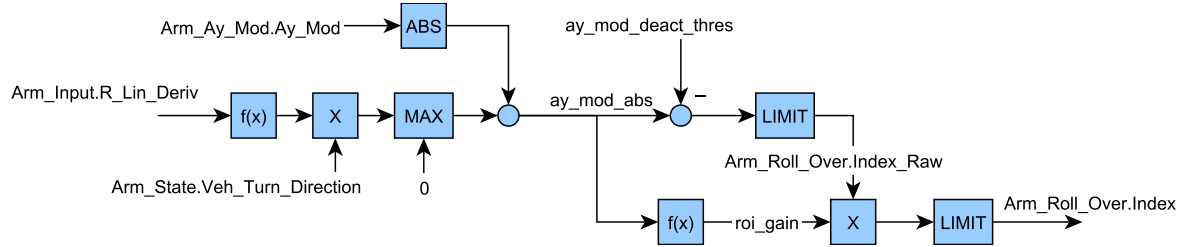


Figure 3.1: Calculation of **ROI**

4 DETAILED DESIGN

This chapter outlines the detailed design of **Arm_Ay_Mod**.

4.1 ALGORITHM DESCRIPTION

4.1.1 **ROI**

ROI is based on the predicted lateral acceleration **AyMod** and can be increased depending on **R_Lin_Deriv**.

$$ay_mod_abs = ABS(AyMod) + f(R_Lin_Deriv) \quad (4.1)$$

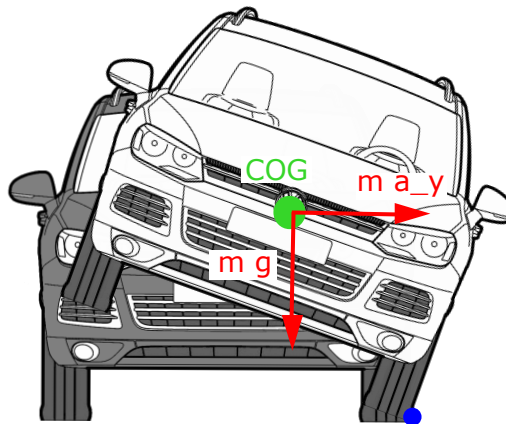


Figure 4.1: Forces on the **COG**

Moments around the contact point (blue circle) yield

$$0 = m a_y \cdot \text{COG_height} - m g \cdot \frac{\text{track_width}}{2} \quad (4.2)$$

The activation and deactivation thresholds are calculated by usage of vehicle dimension.

$$\text{Index_Act_Threshold} = \text{TRIM} \cdot g \cdot \frac{\text{track_width}}{2 \cdot \text{COG_height}} \quad (4.3)$$

$$\text{Index_Deact_Threshold} = \text{TRIM} \cdot g \cdot \frac{\text{track_width}}{2 \cdot \text{COG_height}} \quad (4.4)$$

ROI is the difference between ay_mod_abs and the deactivation threshold.

$$\text{ROI} = (\text{ay_mod_abs} - \text{Index_Deact_Threshold}) \cdot f(\text{ay_mod_abs}) \quad (4.5)$$

Furthermore, ROI can be adjusted by the use of a 2DimensionalLookupTable that is depending on ay_mod_abs. (See Fig. 10.1)

4.1.2 Activation condition

If ay_mod_abs exceeds the activation threshold Index_Act_Threshold, the activation condition Rollindex_Flag is set.

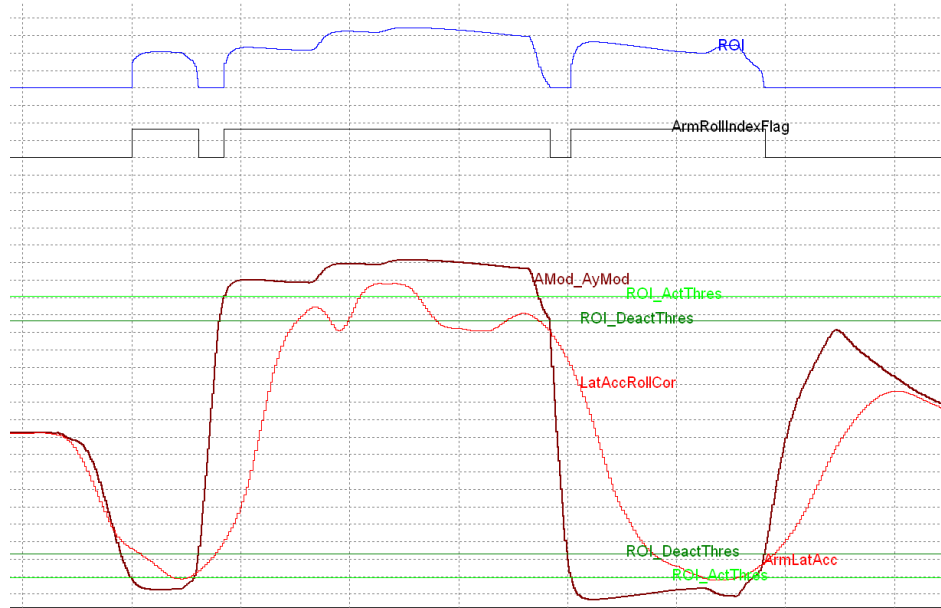


Figure 4.2: Signal flow and resulting Rollindex_Flag

5 PROVIDED TASKS

5.1 ArmRollOverIndex

Description	ArmRollOverIndex
Rate	Standard Arm timebase 5 - 20ms
Dependency	
SYNTAX	void ArmRollOverIndex (void)

6 PROVIDED SERVICES

No provided services.

7 CONFIGURATION OPTIONS

None.

8 CONFIGURATION PROCEDURES

None.

9 TEST CASES

Test category	
Requirements	
Test Case ID	

10 CALIBRATION INFORMATION

10.1 Signal Processing

It is advised to keep the core trimming. If it is necessary to scale the roll index, one can use **ROI_GAIN_TBL**. However, keep in mind to change corresponding thresholds like **ENGINE_ROI_ACT_THRESH** and **ENGINE_ROI_DEACT_HYST** respectively. Since **ARM** pressure request is directly linked to **ROI**, its specific parameters influence directly the pressure request. Due to some calculations this behavior is not obvious. The **ROI_GAIN_TABLE**, which is a function $f(\text{ay_mod_abs})$, is a crucial parameter. Without considering the **Speed_Gain**, it holds:

$$\text{roi_pressure_cmd} = \text{VEH_PRESS_GAIN} \cdot \text{ROI} \quad (10.1)$$

$$= \text{VEH_PRESS_GAIN} \cdot f(\text{ay_mod_abs}) \cdot (\text{ay_mod_abs} - \text{ay_mod_deact_thres}) \quad (10.2)$$

It is more convenient to change the **ROI_GAIN_TABLE** with respect to **Ay_Mod** with known **deact.thres** than with respect to **ay_mod.abs**. The Figure 10.1 below shows a suitable (blue) and an unsuitable (orange) parameter set.

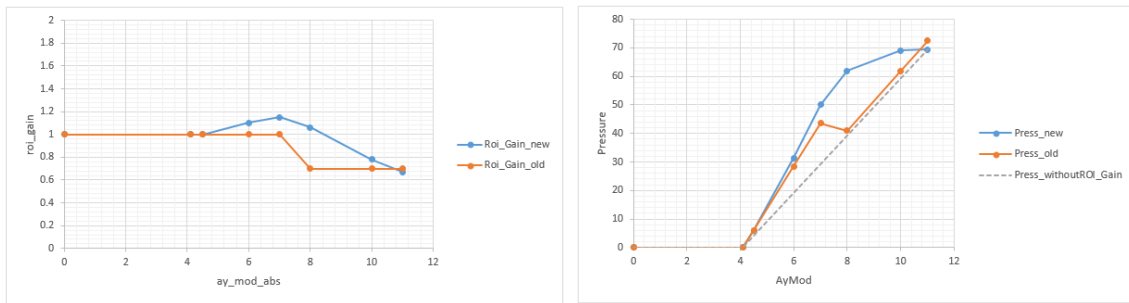


Figure 10.1: Influence of **ROI** on Pressure Command

The shape of **ROI** can be further modified with **ROI_RLIN_DERIV_LEAD**. This parameter is used to inflate the pressure request during lane change when entering the new lane. The driver steering velocity, which is captured in **R_lin_Deriv**, is an indicator to increase the pressure request temporarily. This effect is favorable to slow the vehicle down and increase wheel slip which will lead to a larger turn radius and hence reduce the risk of rolling over. However, since **R_lin_Deriv** is small after the new lane was entered, the pressure request is reduced and faded back to the original pressure request which will ideally avoid a harsh Wheel Slip Regulator (**WSR**) intervention. Bouncing and jumping is often caused by these **WSR** dumps. The pressure inflation can be observed in Fig. 10.2.

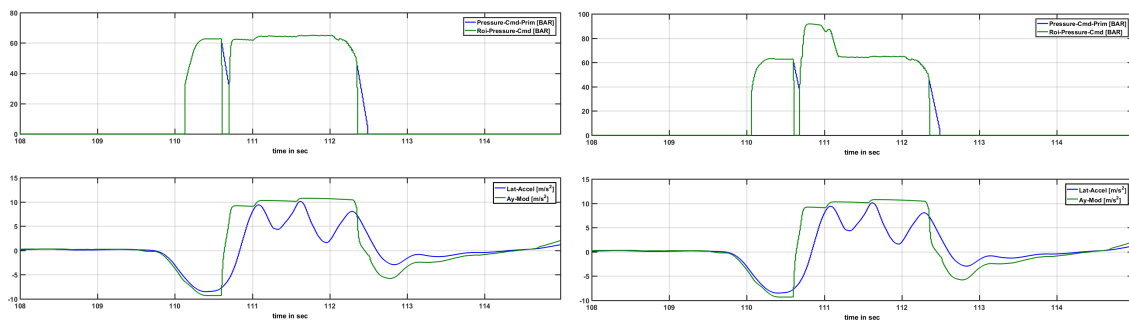


Figure 10.2: Without Inflation (left), With Inflation (right)

10.2 Thresholds

Please refer to ARM-DDD-Arm.State.pdf for activation and deactivation thresholds.

11 APPENDIX